

# ANUPREET SINGH

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## EDUCATION

### University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

### Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

## SKILLS / RELEVANT COURSEWORK

**Programming Languages:** Python, JavaScript/TypeScript, C/C++, Java, SQL

**Tools and Frameworks:** PyTorch, LangChain, CrewAI, Qdrant, Django, FastAPI, React, Next.js, MongoDB, PostgreSQL, PyTest, GoogleTest, AWS, Docker, Kubernetes, GitHub Actions(CI/CD), Claude Code

**Areas of Interest and Coursework:** AI & LLMs, Data Structures & Algorithms, Software Engineering (SDLC), Distributed System Design, Object Oriented Design (OOD), Computer Architecture

## INDUSTRY EXPERIENCE

### Data AI Engineer | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Developed backend services for an AI-native **GRC platform** that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.
- Implemented the **RAG chat agent** with its React interface, marking each control fully, partially, or not satisfied and answering follow-up questions on the unmet ones, with every answer grounded in citations retrieved from the vector index, so no compliance verdict rests on **fabricated control text**.
- Built a priority-ordered **fallback across 5 embedding models** (OpenAI, Anthropic, Ollama), re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising **evidence-backed control assessment accuracy from 64% to 92%**.
- Diagnosed an out-of-memory failure in that fallback path, tracing it to freshly pulled 4-5 GB model weights lingering in the OS page cache, and fixed it with targeted `posix_fadvise` eviction, **eliminating model-load failures**.

### Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built an AI-powered **PRD review platform** in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.
- Architected a fully **serverless review pipeline** on AWS (S3 + CloudFront-hosted React SPA, Django on Lambda, DynamoDB, and an SQS-triggered Lambda AI worker) that retries failed analyses off a dead-letter queue and logs per-report model API spend, sustaining **200+** submissions across 5 cohorts at ~\$3/month idle cost.

### Software Development Engineer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Architected a **self-serve quoting platform** that priced jobs with deterministic multi-stage evaluation, JSONB encoded conditions, per-quote audit trails of every rule applied, and a write-invalidated **Redis** cache, all driven by rules editable from an admin console instead of a developer deploy, cutting quote turnaround from ~**2 days** to **<1 min**.
- Built the **online ordering and booking platform** by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to Kafka workers for order confirmation, customer notification, and crew allocation, taking **\$600K+** in online bookings across ~1,100 orders.
- Delivered the **React SPA** that carried a customer from property details to instant estimate to paid deposit in one client-routed flow, with role-gated admin routes and JWT access tokens backed by `httpOnly` refresh cookies, enabling a streamlined booking flow with ~**40%** of bookings occurring outside office hours without phone support.
- Designed and deployed the **AWS architecture** that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for ~10× seasonal peaks and load-tested to ~**50 quote requests/sec at <120 ms p99** with zero-downtime rolling deploys.

## ACADEMIC EXPERIENCE

### Graduate Assistant | UMBC - Data Structures & Software Engineering

Jan 2025 – June 2026

- Designed over 10 C++ assignments on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for **40+** students.

- Developed **200+** unit tests for the DSA projects on Linked Lists, Self Balancing Trees, Heaps, and Hash Tables automating logic checking and reducing grading time by over **80%**.
- Helped **50+** students design and debug **C++** projects implementing core data structures, and graded their homework.
- Held office hours to clarify technical concepts and translate requirements into actionable implementation steps.
- Mentored 10 student **Scrum** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.

#### Graduate Assistant | *UMBC - Computer Architecture & Software Engineering*

Aug 2025 – June 2026

- Designed a C++ assignment on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for **40+** students.
- Mentored 10 student **Scrum** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.

#### Teaching Assistant | *UMBC - Data Structures*

Jan 2025 – June 2025

- Developed 20+ unit tests for the hash table project, automating logic checking and reducing grading time by **80%**.
- Helped **50+** students design and debug C++ projects implementing core data structures and graded homework.

## PROJECTS

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### Role Radar | *MacOS app for the latest job-posting alerts*

- Built an async Python **job-ingestion engine** that tracks **7,700+** employers' careers pages, auto-selecting 1 of 22 source-specific scrapers (Greenhouse, Workday, Oracle HCM, ...) by URL and routing every request through a robots.txt-compliant client with per-host rate limits and Retry-After backoff, checking **~3,600 companies per 18-minute pass** across ~1,000 hosts at a **0.18%** request failure rate.
- Designed a **laptop-to-Lambda failover** that keeps checks running while the Mac sleeps, electing a single runner through a conditionally written DynamoDB lease whose epoch acts as a **fencing token** on each company's transactional save, so a laptop waking mid-run can never overwrite its successor's state, with Lambda taking over within **8 min** and handing back in **~15 s**.
- Designed a native **SwiftUI** macOS app with a guided setup and live job-match tracking that runs from the Dock or menu bar and supervises the checker **daemon** as a **launchd** agent, restarting it within a minute if it stops, so non-technical users get alerts with **no terminal or config files**.

### Rendering Pipelines from Scratch | *Ray tracer, rasterizer, and real-time OpenGL renderer in C++*

- Built a **C++ ray tracer** that parses Polyray scenes and shades them with Blinn-Phong direct lighting, recursive reflection, and Schlick-Fresnel refraction, then accelerated it with a median-split **BVH** over primitive centroids and an atomic row-chunk thread pool, cutting a 512×512 render from **106 s to 0.14 s (~759×)**.
- Engineered a **software rasterizer** carrying triangles through the full world-camera-projection-NDC-raster transform chain with barycentric incremental scan conversion, z-buffering, and back-face culling, extended with Sutherland-Hodgman frustum clipping and **cube shadow maps** that pick 1 of 6 depth faces by dominant axis, rendering omnidirectional shadows no single 90° map can cover.
- Implemented **DXT1 texture compression** end to end in an OpenGL scene renderer, fitting each 4×4 texel block's palette by principal-axis covariance, packing 2-bit selectors into the 64-bit block layout, and compressing every CPU-generated mip level independently, shrinking color textures **6× in GPU memory** behind a runtime A/B toggle.

### Bloodborne TAG | *Text Adventure Game with natural language command parsing*

- Engineered a **command parser** using few-shot Mistral-7B for slot extraction and embedding-similarity-based intent matching, mapping free-form playtest commands to 14 deterministic actions at **96%** intent accuracy.
- Architected a data-driven game engine that loads a **41-room** world and **60+** items/NPCs from JSON into in-memory session state, gating progression on generic key-item fields so new content ships with **zero** engine code changes.

### When Pixels Talk Back | *Watermarking EMR Data to analyse impact on CV models*

- Fine-tuned **3 computer-vision models** (ResNet50, DenseNet169, MobileNetV2) for classification over COVID-19 lung CT scans via two-stage ImageNet transfer learning, to establish a baseline peaking at **93%** accuracy and **0.951 F1**.
- Embedded a 1024-bit blind fragile watermark into the Region-of-Non-Interest of CT scans via key-scrambled LSB embedding of a SHA-512 fingerprint, enabling content authentication with near-lossless quality at **SSIM 0.999**.
- Benchmarked 3 watermarking algorithms across all 3 models, showing the fragile scheme costs **0%** accuracy loss, versus ~1% for reversible embedding with a 300-bit payload and 11.2% for robust embedding.

### Nothing Lost in Translation | *Lexically constrained Sequence Generation*

- Reconstructed a beam search decoder from scratch, outperforming **HuggingFace's** stock `.generate()` by 1.45 BLEU.
- Built a **Grid Beam Search decoder** that guarantees positive/negative lexical constraints in any autoregressive sequence generation, demonstrated on English-to-Russian NMT with a **15-point BLEU** gain (25.45 → 40.59).